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Aftertreatment Diesel Exhaust Fluid (DEF) Dosing and Selective Catalytic Reduction (SCR) System Reuse Guidelines

Aftertreatment Diesel Exhaust Fluid (DEF) Dosing and Selective Catalytic Reduction (SCR) System Reuse Guidelines

This document provides reuse guidelines for the aftertreatment diesel exhaust fluid (DEF) dosing and selective catalytic reduction (SCR) systems, in conjunction with the information provided in the corresponding Service Manual.

Product Affected

- ISB4.5 CM2350 B104
- ISB6.7 CM2250
- ISB6.7 CM2350 B101
- ISB6.7 CM2350 B103
- ISC8.3 CM2250
- ISL9 CM2250
- ISF2.8 CM2220
- ISF2.8 CM2220 AN
- ISF2.8 CM2220 IAN
- ISF3.8 CM2220 AN
- ISF3.8 CM2220 IAN
- ISL CM2350 L101
- ISX12 CM2350 X102
- ISX12/ISX11.9 CM2250
- ISX15 CM2250
- ISX15 CM2350 X101
- QSB6.7 CM2350 B105
- QSF3.8 CM2350 F107
- QSK19 CM2450 K155G
- QSL9 CM2350 L102

The reuse guidelines contained in this document are **only** Cummins Inc. approved inspection methods and pass/fail criteria with regards to inspecting an aftertreatment SCR catalyst.

The aftertreatment SCR catalyst housing **must** be free of dents. Mounting flanges **must** be free of dents, cracks, or gouges to seal properly with gaskets.

Replace any oil, fuel, or coolant-soaked aftertreatment SCR catalyst.

Note : The aftertreatment DEF dosing unit filter is a non-reusable maintenance item. The information below is provided for use in determining if contaminated DEF has passed through the dosing system.

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Obsolete Filter



07900597

Current Pleated Filter

Aftertreatment DEF Dosing Unit Main Filter - New

Appearance:

Clean filter element



11b00112



07900598

Aftertreatment DEF Dosing Unit Filter - Normal Use - Approximately 200k Miles	
Appearance:	Darkened filter element, urea odor, absence of large particles, absence of hydrocarbon odor
Cause:	Normal dirt and dust intrusion into DEF system
Action:	Replace DEF dosing filter at normal maintenance interval.



Aftertreatment DEF Dosing Unit Filter - Contaminated	
Appearance:	Dark yellow filter element (similar appearance on both style filters)
Cause:	Likely diesel fuel in the DEF tank

Aftertreatment DEF Dosing Unit Filter - Contaminated

Action:

Remove and clean DEF tank, flush DEF lines, and replace DEF dosing unit filter. Reference the following procedure for contamination/incorrect fluid:

- Use the following procedure in the ISF2.8 CM2220, ISF2.8 CM2220 EC, ISF2.8 CM2220 AN, ISF2.8 CM2220 IAN Service Manual, Bulletin 4022178. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/31/31-011-056-tr-isb07.html)
- Use the following procedure in the ISF3.8 CM2220, ISF3.8 CM2220 AN, and ISF3.8 CM2220 IAN Service Manual, Bulletin 4021704. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/118/118-011-056.html)
- Use the following procedure in the ISB6.7 CM2250 Service Manual, Bulletin 4022254. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/31/31-011-056-tr-isb10.html)
- Use the following procedure in the ISC8.3 and ISL9 CM2250 Service Manual, Bulletin 4022257. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/45/45-011-056-tr-iscl10.html)
- Use the following procedure in the ISX12/ISX11.9 CM2250 Service Manual, Bulletin 2883445. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/132/132-011-056-tr-vqxi6.html)
- Use the following procedure in the ISX15 CM2250 Service Manual, Bulletin 4022250. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/132/132-011-056-tr-mixar5.html)
- Use the following procedure in the QSB6.7 CM2350 B105 Service Manual,

Aftertreatment DEF Dosing Unit Filter - Contaminated

Bulletin 4332778. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/302/302-011-056.html)

- Use the following procedure in the QSL9 CM2350 L102 Service Manual, Bulletin 4332796. Refer to Procedure 011-056 in Section 11

(/qs3/pubsys2/xml/en/procedures/302/302-011-056.html).

- Use the following procedure in the ISB6.7 CM2350 B101 Service Manual, Bulletin 2883567. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/155/155-011-056-tr-isb13.html)

- Use the following procedure in the ISL CM2350 L101 Service Manual, Bulletin 4310787. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/185/185-011-056-tr.html)

- Use the following procedure in the ISX12 CM2350 X102 Service Manual, Bulletin 4310646. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/174/174-011-056-tr-pbix13.html)

- Use the following procedure in the ISX15 CM2350 X101 Service Manual, Bulletin 4310641. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/173/173-011-056-tr-prix13.html)

- Use the following procedure in the ISB6.7 CM2350 B103 Service Manual, Bulletin 4332641. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/194/194-011-056-tr.html)

- Use the following procedure in the ISB4.5 CM2350 B104 Service Manual, Bulletin 4332646. Refer to Procedure 011-056 in Section 11.

Aftertreatment DEF Dosing Unit Filter - Contaminated

(/qs3/pubsys2/xml/en/procedures/186/186-011-056-tr.html)

- Use the following procedure in the QSF3.8 CM2350 F107 Service Manual, Bulletin 4367316. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/279/279-011-056.html)

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Aftertreatment DEF Dosing Unit Filter - Contaminated

Appearance:

Grease-like substance separated from DEF
(similar appearance on both filters)

Cause:

Contaminant in DEF tank

Aftertreatment DEF Dosing Unit Filter - Contaminated

Action:

Remove and clean DEF tank, flush DEF lines, and replace DEF dosing unit filter. Reference the following procedures for contamination/incorrect fluid:

- Use the following procedure in the ISF2.8 CM2220, ISF2.8 CM2220 EC, ISF2.8 CM2220 AN, ISF2.8 CM2220 IAN Service Manual, Bulletin 4022178. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/31/31-011-056-tr-isb07.html)
- Use the following procedure in the ISF3.8 CM2220, ISF3.8 CM2220 AN, and ISF3.8 CM2220 IAN Service Manual, Bulletin 4021704. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/118/118-011-056.html)
- Use the following procedure in the ISB6.7 CM2250 Service Manual, Bulletin 4022254. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/31/31-011-056-tr-isb10.html)
- Use the following procedure in the ISC8.3 and ISL9 CM2250 Service Manual, Bulletin 4022257. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/45/45-011-056-tr-iscl10.html)
- Use the following procedure in the ISX12/ISX11.9 CM2250 Service Manual, Bulletin 2883445. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/132/132-011-056-tr-vqxi6.html)
- Use the following procedure in the ISX15 CM2250 Service Manual, Bulletin 4022250. Refer to Procedure 011-056 in Section 11.
(/qs3/pubsys2/xml/en/procedures/132/132-011-056-tr-mixar5.html)
- Use the following procedure in the QSB6.7 CM2350 Service Manual, Bulletin

Aftertreatment DEF Dosing Unit Filter - Contaminated

4332778. Refer to Procedure 011-056 in Section 11.

(/qs3/pubsys2/xml/en/procedures/302/302-011-056.html)

- Use the following procedure in the QSL9 CM2350 Service Manual, Bulletin 4332796. Refer to Procedure 011-056 in Section 11
(/qs3/pubsys2/xml/en/procedures/302/302-011-056.html).
- Use the following procedure in the ISB6.7 CM2350 B101 Service Manual, Bulletin 2883567. Refer to Procedure 011-056 in Section 11.
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- Use the following procedure in the ISL CM2350 L101 Service Manual, Bulletin 4310787. Refer to Procedure 011-056 in Section 11.
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Document History

Date	Details
2012-7-24	Module Created
2013-10-23	none
2013-11-26	Update Engine Model Names
2014-9-3	none
2018-6-8	Added Product Affected Section.
2024-2-27	Added QSK 19 CM2450 K155G.